

# Developing the access network

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*An overview is presented of current and forthcoming access network technologies and how they are likely to be deployed in access networks around the globe. These technologies are discussed in the light of commercial considerations, such as the current market structure, the impact of competition and regulation, as well as the demand for new broadband and Internet services. Various technology options are introduced and discussed, and issues presented that would influence the choice of such technologies for different operating companies.*

## 1. Introduction

Fixed access networks have always been a critical component of telecommunications networks. Essentially they provide the final connection through to the customer and at the same time they are usually the most expensive component in terms of capital investment and ongoing cost of maintenance and repair — typically they may form some 70% of the total investment required. Competition in the access and local telecommunications market has become fierce in recent times, as major players seek to form new mergers and acquisitions, and cable TV companies and new entrants compete head-on with traditional telcos (PTTs). This competition is being fuelled by regulatory changes as markets become more liberalised, and indeed by new technology which increasingly threatens to make existing operators and their networks obsolete. Already there has been a massive investment into fixed access networks around the globe where some 700 million access lines have now been installed — by far the majority using twisted copper pairs. Increasingly, new access technologies are being developed and deployed with a view to delivering new services and to reducing capital and operating costs. However, with such a large installed base of twisted pairs, a key focus of the new technologies has also been to enhance the capability of existing copper.

There are a considerable number of technology options available to both existing and new operators, and the optimum choice will be dependent on a number of different factors. Some of the technology options are discussed in this paper (and in more detail in subsequent papers in this Journal) and factors determining the choice of such options will be presented.

### 1.1 Definition of access

Figure 1 shows the key elements of access networks as used today. Traditionally the access network connected the

customer through to the local exchange (or central office) and did not include exchange equipment. The only part of exchange equipment included was the main distribution frame — the first point of flexibility in the network. More recent definitions, however, separate out the aggregation business (access) from the transport business (core). Aggregation equipment would include the fixed access connection to the customer's premises (copper, fibre or radio) as well as any aggregation devices such as multiplexers; in particular, it also includes exchange line cards (digital line interface into switch). The new definition takes into account the impact of new fibre access technologies, which can shift the position of line cards out to the customer's premises or into the street, as well as providing concentration, although this can still be categorised as aggregation.

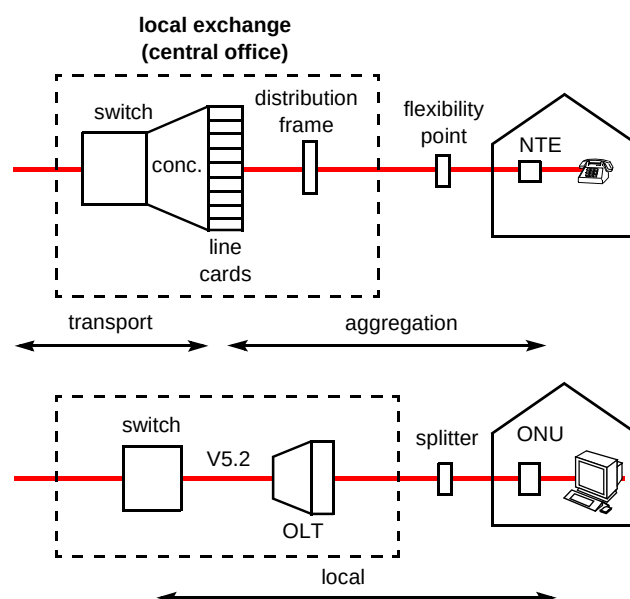


Fig 1 Definition of access network.

The market in the USA is categorised slightly differently into local and long distance, with local networks in this case including the local switch and providing an end-to-end call connection between customers.

## 1.2 UK market structure

The liberalisation of the market in recent years has led to some 200 new operating licences being granted in the UK, offering a range of different services. Figure 2 is an attempt to classify some of the key operators in terms of their services and the customers they are targeting.

The traditional telco is shown offering telephony services to all customers, and Mercury (MCL) shown in its pre-merger form. The diagram gives a view of the number of niche operators that have been formed in recent years, some competing with the PTTs and others, such as the cable TV companies, also offering new services. By and large different companies have selected different access technologies and Table 1 gives a view of technologies used today. These technologies are summarised in section 2.

Table 1 Access technologies.

| Services                        | Access network     |
|---------------------------------|--------------------|
| Bulk telephony/private circuits | Copper pairs/fibre |
| Mobile/niche telephony          | PCN/fixed radio    |
| Metropolitan/business services  | Fibre access/SDH   |
| Broadcast TV/telephony          | HFC/copper pairs   |
| Broadcast TV only               | Satellite          |
| Long distance/Internet          | Indirect access    |
| Value-added service providers   | Leased lines       |

## 2. Current networks

### 2.1 Copper access network

An outline structure of BT's telephony network is given in Fig 3. Over many decades BT's access network has grown to encompass access nodes in over 5600 buildings and an overhead network comprising nearly 1.5 million poles. This network reaches all corners of the UK and is capable of delivering services to the vast majority of dwellings and businesses.

In the main, access is provided by copper pairs with some 28 million working pairs in the UK today terminating on a two-tier switching network, comprising a small number of trunk switches (DMSUs) used to interconnect a large number of local switches (DLEs). The MDF is the main point of physical flexibility in the exchange allowing copper pairs to be flexibly connected into the switch. E-side (exchange) cables might typically consist of 1000 pairs and fan out from exchanges to PCPs — the main flexibility points in the network. A PCP typically serves some 350 customers and allows better utilisation of E-side cables. D-side (distribution) cables might typically consist of some 100 pairs and fan out from PCPs to DPs. DPs are the final flexibility points in the network and typically serve about 12 customers. Although E-side cables are usually installed underground, D-side cable exists both underground and overhead. In the UK, duct is used to allow additional cables to be installed cost effectively as required. In continental Europe access cables frequently are directly buried.

Because of the uncertainty in demand for service, and the desire to keep connection times down to a minimum, a considerable number of spare pairs have to be installed,

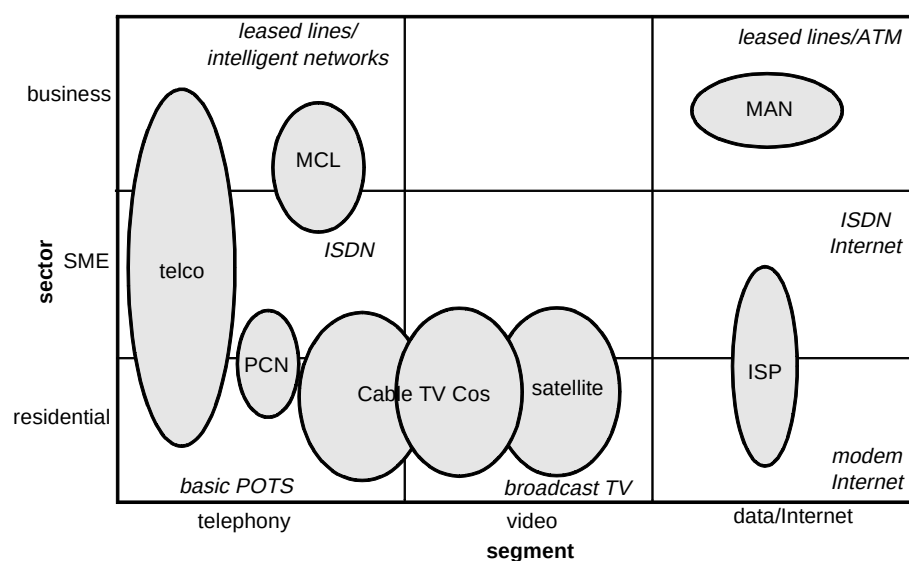


Fig 2 UK market structure.

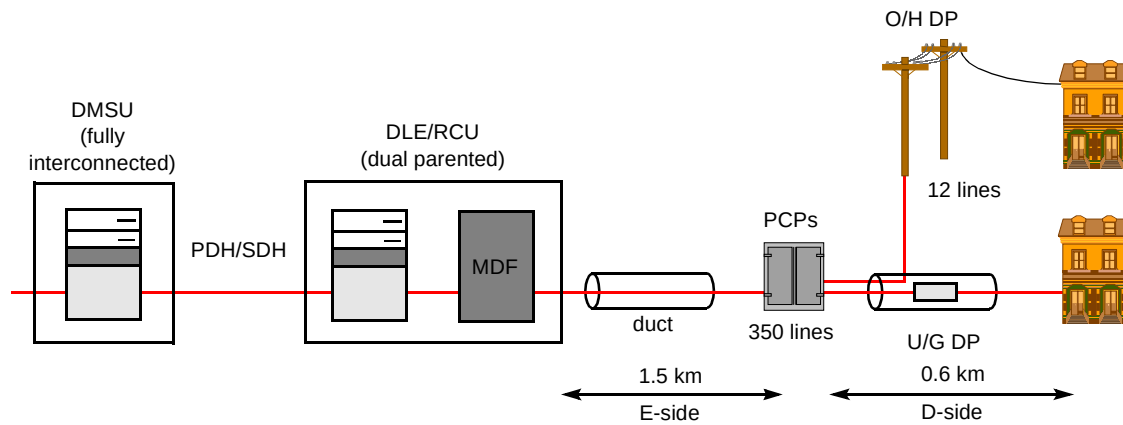


Fig 3 Copper access network.

based on telco planning rules. Directly buried networks generally need to have more spare capacity installed than ducted cable. Typical distances are given in Fig 3, and are characteristic for Europe where population densities tend to be high.

Not surprisingly, the cost of such a network is dominated by fixed costs resulting from depreciation and overheads such as accommodation. However, as a result of many initiatives in recent years direct operating costs, although substantial, are now among the world best.

Although access networks have traditionally been vulnerable to the effects of weather (e.g. storm damage or water ingress), as a result of continuous improvement, the average customer will now see just one fault in seven years. There is still scope for improving the weather resilience of the access network, since the unpredictability of severe weather leads to less than optimum deployment of labour resources. Detailed studies show that modern access plant is inherently reliable and that intervention and work practices drive fault rates.

## 2.2 Cable TV network

Regulatory freedom in the UK has encouraged cable TV operators to roll out new networks offering both cable TV and telephony services. The typical network shown in Fig 4 indicates how the two services are offered. Although technology now exists to offer both TV and telephony services over a single integrated co-axial cable network, at the time these networks were installed a safer option was to install both co-axial cable and copper pairs for the final connection. Optical fibre was used as the E-side cable.

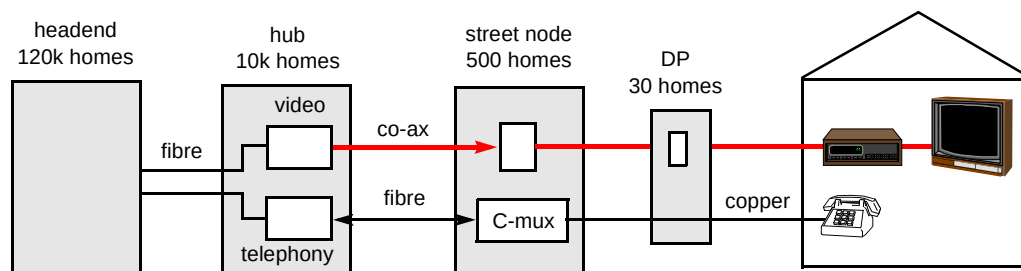


Fig 4 Cable TV network.

Essentially there are two parallel networks side by side — hybrid fibre/coaxial cable for the TV network and hybrid fibre/copper pairs for telephony. A primary multiplexer (C-mux) is installed in the street node so that signals on copper pairs can be digitally multiplexed on to optical fibre. The networks are integrated at the physical cable and duct level which is where most of the cost saving can be found. Local authorities generally require new cable TV networks to be installed entirely underground using duct. Analogue TV is distributed using AM-VSB, although two-way digital upgrades are now being installed across the UK.

## 2.3 Other current networks

Other niche operators are now offering services and usually specialise in only one of the many types of access technology available. These are discussed in section 4 but are summarised here.

- Mobile operators use GSM or other PCN technology. In addition, a number of companies provide fixed access using point-to-multipoint radio equipment over limited geographic areas.
- Direct fibre through to customers is now increasingly used both by BT and other operators, although because of the high cost associated with civil works it can only really be justified for high bandwidth connections to large business customers. These point-to-point fibre systems generally use either PDH or, increasingly, SDH multiplexers and can provide bandwidths from 2 up to about 155 Mbit/s.

- Satellite technology can be used effectively to distribute broadcast TV and indeed BSkyB has by far the largest number of connections in the UK.

At this stage it is also worth mentioning that the lack of an access network does not preclude operators from offering services to customers. Indirect access, through the use of another operator's network, allows long-distance operators to provide service, and indeed this is the norm in the USA market. Indirect access is also used by Internet service providers to allow dial-up connections into their Internet services.

### 3. Market dynamics

The choice of access technology is crucial to the survival of operators since access is such a dominant part of overall costs. That choice will, however, be dependent on a number of factors, best summarised through the use of the Porter diagram of competitive forces (Fig 5). New entrants have been discussed in section 2.2, with many new niche operators selecting specific technology to support their overall strategy. The new technologies available are discussed in section 4.

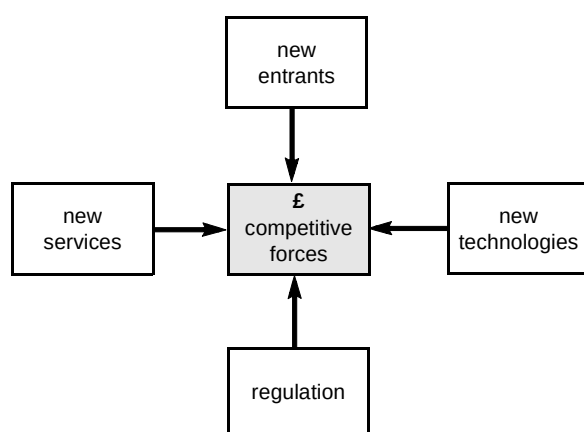


Fig 5 Competitive forces.

#### 3.1 Regulation

Regulation without doubt makes a significant contribution to the choice of network. Competition was first introduced in the UK in 1984 when a duopoly was established allowing both BT and then MCL to offer services. (This followed liberalisation within the USA and divestiture of AT&T in 1984, as well as the modified final judgement giving RBOCs the exclusive right to offer local services and allowing competing long-distance companies.) Following the duopoly review in 1991, the UK market was opened to much more competition, with the cable TV companies allowed to offer full telephony services.

entrants were encouraged into the market-place through the implementation of favourable rules; for example, BT was prohibited from offering broadcast TV services and from the general use of fixed radio technologies. Regulation was used to encourage new access infrastructure build within the UK and the policy was largely successful. This was based on the assumption that economies of competition (through competing networks) outweighed economies of scope — whereby a single network may have been used to offer competing services.

As the market has matured regulatory policy has shifted towards 'behavioural regulation' whereby established players are discouraged from anti-competitive behaviour through the use of penalties. In addition, operators are encouraged to move towards more open networks, and network interconnection established on fair terms. One contentious area is 'copper unbundling' whereby operators may lease the use of copper pairs from another operator to offer their own services. Inevitably this would entail collocation of equipment since interconnection at the MDF would require additional equipment in the exchange. Another contentious area is 'infrastructure sharing' whereby operators may share the use of physical assets such as duct and cable. Although not aggressively pursued in the UK yet, it seems likely that, as we move towards a more mature market with full open competition, operators will inevitably co-operate as well as compete in order to prosper.

#### 3.2 New services

A key factor in determining the choice of technology is of course the services that will be offered over the network. Undoubtedly, they will broaden out from the three key services of today — telephony, cable TV and mobility — to include:

- teleworking,
- video-on-demand,
- home shopping,
- tele-education,
- videoconferencing,
- LAN interconnection,
- intranet,
- high-speed Internet.

Indeed these new services will begin to dominate over a period of time in terms of overall revenues, as margins on commodity services reduce. Figure 6 is one view of how the overall market mix for services may change within the UK over a period of ten years. The figure has been compiled

from a variety of sources and is meant for illustrative purposes only.

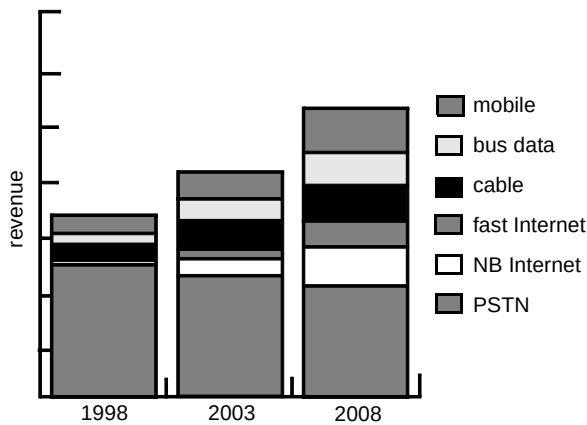


Fig 6 The market mix is changing.

Although to a certain extent new services will be offered at low bandwidths (9.6 to 56 kbit/s) feasible with V32 modems over conventional copper pairs, the requirement for video services means that customers will be better served by higher speed digital access connections.

Some outline requirements in terms of bandwidths for these new services are given in Table 2.

Table 2 Digital access requirements.

|             | Downstream/upstream data rates                        |
|-------------|-------------------------------------------------------|
| Corporate   | LAN interconnect, intranet (34, 155 Mbit/s symmetric) |
| Teleworking | Remote LAN access (0.5/0.5 Mbit/s, 2.0/0.5 Mbit/s)    |
| SME         | Fast Internet access (0.5/0.5 Mbit/s, 2.0/0.5 Mbit/s) |
| Residential | Fast Internet access 1999 (0.5/0.5 Mbit/s)            |
|             | Entertainment 2000 (VOD 2.0/0.5 Mbit/s)               |

#### 4. New and emerging access technologies

The delivery of innovative services requires the presentation of new and increasingly digital interfaces at the network termination. In many cases, the capacity required is in excess of those required for single-line PCM speech channels (64 kbit/s). This has led to a range of developments in access technologies which give network operators a great variety of options through which customers may be provided with services.

The selection of technology depends not just on the service requirements of any one customer, but also on the type of network provider and the access infrastructure already in place (see Table 3).

Table 3 Access options.

| Type of operator                         | Example of current access infrastructure                                        | Most appropriate new technologies                                                                                  |
|------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Incumbent operator in developed country  | Twisted pair copper dominates (high penetration), but substantial optical fibre | xDSL on copper, expansion of fibre (PON + point-to-point) in existing duct                                         |
| Cable TV operator                        | Coaxial cable, some optical fibre                                               | Cable modems (if possible)                                                                                         |
| Incumbent operator in developing country | Twisted pair copper (but low penetration)                                       | Wireless local loop, expansion of fibre (PON + point-to-point)                                                     |
| New entrant in developed country         | None                                                                            | Mixture of fibre, point-to-point radio, wireless local loop, leased lines from other operators and indirect access |
| New entrant in developing country        | None                                                                            | Wireless local loop, satellite systems for quick roll-out, leased lines and indirect access                        |

A schematic of a selection of the access transport options available to current network operators is shown in Fig 7. In this example, the access network is based on an SDH fibre ring, which provides aggregation of traffic on to protected paths and a high degree of resilience between the service switches/routers and the access add/drop multiplexers (ADMs). From the ADMs, connections to customer sites are provided over star or tree-and-branch networks.

Technological innovations are being matched by an increasing level of network management functionality in the access network, in certain cases (particularly where customer SDH/SONET is provided). This can allow the management of core and access networks from a common network control layer, removing, to a certain extent, the boundary between core and access networks.

The choice of technology with which to serve a particular customer or group of customers depends on a number of parameters, the key issue being the aggregate bandwidth to be delivered to/from the customer site. Where a network operator already possesses access infrastructure that enters the customer premises, the approach should be to provide services over existing infrastructure if at all possible. The focus is on the enhancement of capabilities by changing only the equipment at either end of the access network, rather than making wholesale changes to the access infrastructure. This approach enables network operators to reduce both the time and the cost of responding to customer requirements.

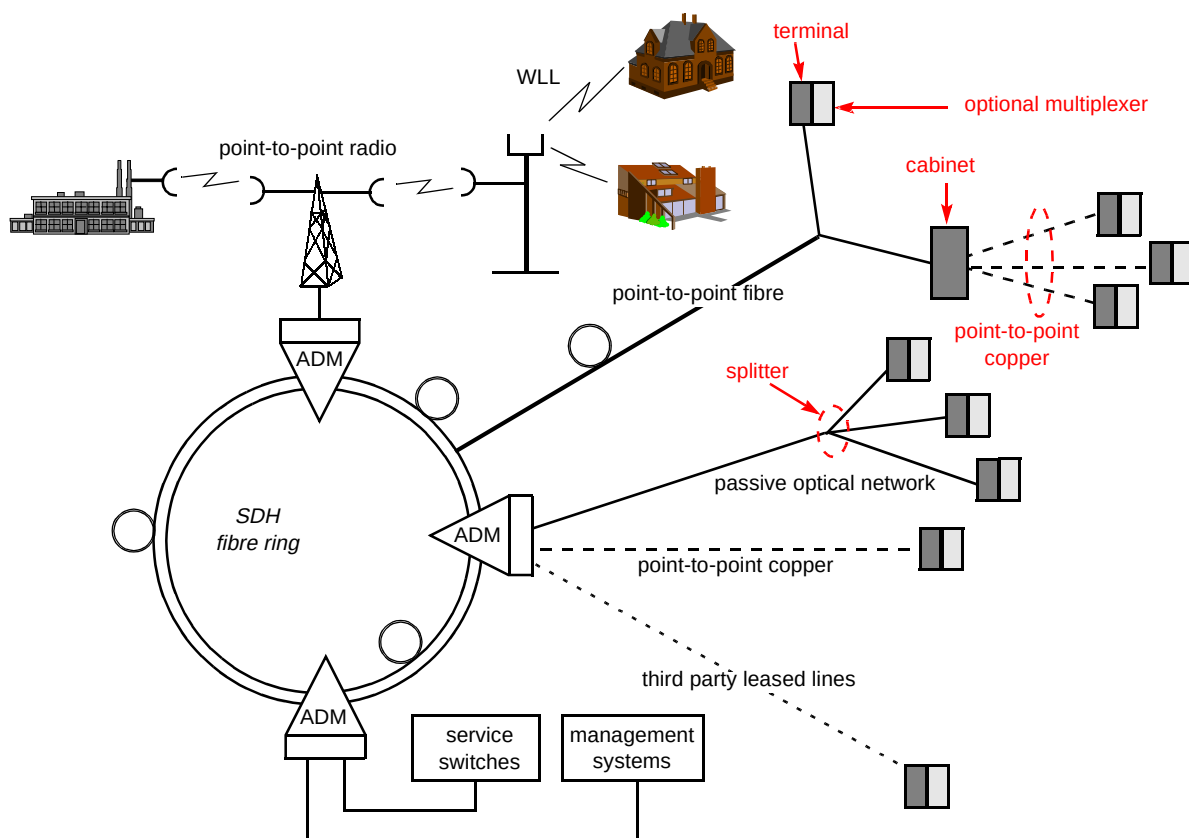


Fig 7 Modern access networks.

New network operators may choose to obtain customer access using incumbent (or other competitor) facilities such as interconnect, leased lines, or 'dark copper' (copper unbundling), where regulation permits. All these options require a substantial outflow of current account payments to competitors. Consequently, there are a growing number of new operators who are, to some degree, deploying their own access infrastructure. In addition to bandwidth, the choice of technology to serve customers depends upon the proximity to other customers or potential customers — a concentrated customer cluster could be most efficiently served by optical fibre infrastructure for high bandwidths or wireless local loop (WLL) for telephony. For incumbent and new network operators, efficient planning systems and forecasting techniques are essential to minimise the impact of changed or new customer requirements on the access network infrastructure.

The sub-sections that follow provide a summary of key access technologies. A more detailed overview of how these technologies are currently deployed in the BT UK network are given in Adams et al [1].

#### 4.1 Wireline technologies

##### Twisted pair copper cables

The rapid growth in dial-up access to the Internet has led to customers using 'high-speed' voiceband data modems which are capable of operation at data rates of up

to 33.6 kbit/s over analogue copper lines. Most recently, asymmetric modems capable of operation at data rates of up to 56 kbit/s have become commercially available. However, this technology is reaching the fundamental limits set by Shannon's Law for transmission using these types of protocol over installed twisted pair cables.

Further enhancements to data rates on copper cables require the use of different formats of line coding, implemented by the connection of terminals to the main distribution frame (MDF) in two main classes — integrated services digital network (ISDN) basic rate at 128+16 kbit/s and secondly, the range of standard methods known generically as digital subscriber line (DSL). There are a range of DSL systems operating at a variety of data rates which are often referred to xDSL (Fig 8) [2]. The most widely used type is high-speed DSL (HDSL), which provides 2 Mbit/s transmission over distances of up to several kilometres. Transmission spans may be increased by using multiple pairs. Increasing data rates may be provided by techniques such as asymmetric DSL (ADSL — up to 6 Mbit/s towards the customer) and very high speed DSL (VDSL — typically 12 Mbit/s), although increases in data rate are obtained at the expense of reductions in the transmission distances.

##### Co-axial copper cables

The dominant usage of co-axial copper cables in access networks has been to provide the bearer for analogue cable

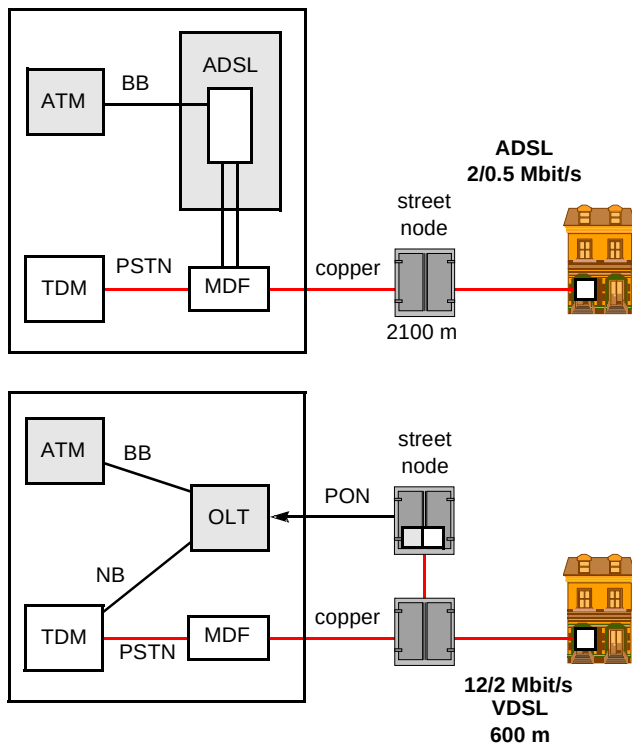


Fig 8 DSL technologies for broadband (BB) services.

TV networks. Where the existing allocation of RF spectrum to the broadcast TV signals permits, unused spectrum may be used for data transmission between terminals known as 'cable modems' (640 kbit/s is typical). New cable TV network architectures, particularly hybrid fibre/co-ax (HFC) (see Fig 9) are being deployed [3]. The final customer connection is over a much shorter co-axial cable than the original 'all-co-ax' network, implying that end-to-end transmission at a higher data rate will be possible with this new architecture.

#### Electrical mains power cables

Telecommunications deregulation in many countries has led to the emergence of network operators with a wide

range of owners. In particular, electricity distribution companies have diversified into telecommunications, in many cases using their regional or national grids as the basis of a core network. To enable extensions into the access area, new technology has recently become available to carry digital signals (up to a shared 1 Mbit/s) over the electrical mains distribution network. This technology is being trialled within the UK, although key concerns are regarding both the capacity available to a customer and radio frequency interference [4].

#### Optical fibre cables

Optical fibre cable systems are being deployed in access networks to carry aggregated traffic in one of two configurations — point-to-point or 'passive optical networks' (PONs). Point-to-point capacities of  $4 \times 2$  Mbit/s are common, although 34 Mbit/s and 155 Mbit/s are also deployed. Figure 10 shows an example configuration of TPON systems that are in established commercial use for fibre-to-the-kerb (FTTK) and fibre-to-the-building (FTTB) applications. TPON systems can provide capacities of up to  $16 \times 2$  Mbit/s and offer a cost-effective alternative to point-to-point configurations for medium-sized sources of traffic. PONs offering ATM-format transmission capacities up to 622 Mbit/s are under development by several suppliers, and have been trialled by a number of network operators worldwide.

#### 4.2 Wireless technologies

##### Point-to-point radio

Point-to-point radio systems are commercially available in a wide range of spectral bands, providing capacity between 2 Mbit/s and multiples of 155 Mbit/s. Radio is particularly suited to applications where customers with bandwidths of 2 Mbit/s or more are situated an appreciable distance from cable networks. The reach of a radio system will be determined by the characteristics of the transmitter/

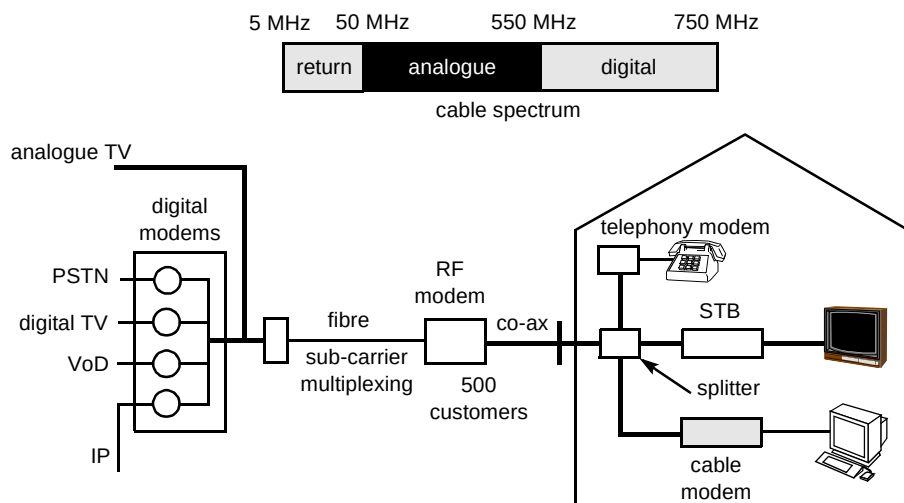


Fig 9 Digital HFC network.

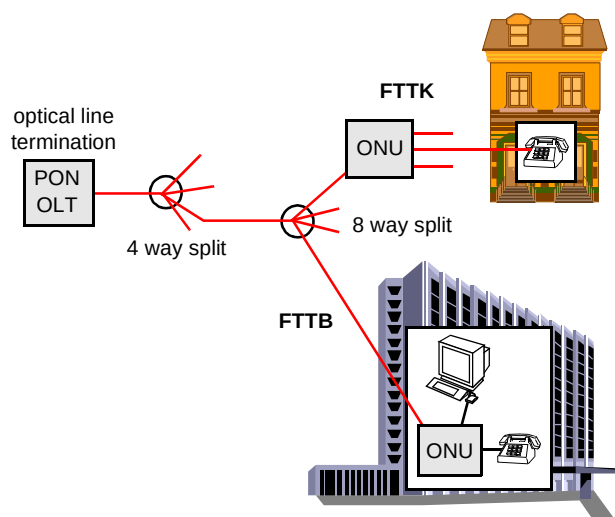


Fig 10 Telephony over passive optical networks (TPONs).

receiver pair and radio propagation in the spectral band(s) available to the network operator.

#### Point-to-multipoint terrestrial radio

Cellular networks provide the most common form of mobile access, with radio channels being assigned to establish connections to customers requiring service. While the service capability of current mobile networks is extensive, the maximum bandwidth available to an individual customer remains somewhat limited. Wireless local loop is a radio alternative to twisted pair copper networks. These systems provide point-to-multipoint radio transmission from a base-station to a fixed network termination [5]. In current commercial products, an  $n \times 32$  kbit/s link is provided to each customer on demand ( $n$  is 1 or 2). Speech paths are provided over 32 kbit/s digital compressed speech channels (adaptive differential PCM, or ADPCM). Two main product groupings currently exist — the first is the digital European standard for cordless telephony (DECT) in the 800 MHz band, where the cell size is typically up to 1 km, with some proprietary systems operating at high frequencies, hence providing larger cell radii (several kilometres is typical). Evolution of these two classes of system is occurring in different directions — DECT towards some limited mobility (within the same base-station area), and the larger cell systems towards ISDN basic rate capability. Convergence of DECT and cellular technology (WLL for 'home' use, GSM for roaming away from home) is already taking place, with dual mode DECT/GSM handsets in commercial production.

A limited number of manufacturers are producing broadband (multiple 2 Mbit/s per customer) point-to-multipoint radio which is being trialled in a number of locations around the world. Just as for WLL systems, bandwidth is assigned to customers on demand, even if the capacity required changes during a period of

communication over the link. However, this is a separate technology to WLL and is not an evolution path from it.

#### Satellite

Satellite technologies are particularly suitable for delivering services over very wide geographic areas. Services that are most appropriately delivered over satellite include:

- subscription-based and pay-per-view broadcast TV, for both residential and business applications,
- contribution quality TV links, particularly from transportable earth stations for occasional use,
- data, particularly using very small aperture terminals (VSATs) in areas where terrestrial access is not available,
- speech, fax and voice-band data with global reach (currently provided via geostationary satellites, but soon to be augmented by several constellations of low and medium Earth orbit satellites under construction).

All of these application areas are undergoing major changes introduced by technological advances [6], including increased bandwidth/decreased cost per bit, low earth orbit satellite mobile systems and digital broadcast TV. New services are becoming available, particularly those which are based on the Internet protocol (e.g. Internet push/pull) and are generally not particularly delay sensitive. Satellite remains an expensive technology, so there is growing interest in high-altitude platforms [4].

#### 4.3 Infrastructure-independent technologies

To complete the portfolio of access options available to network operators, it is necessary to introduce other options which are particularly (though not exclusively) applicable to new operators. Around the world, operators and bandwidth suppliers lease capacity that can provide access links. To increase the range of services that can be carried over a single leased circuit, equipment suppliers produce a range of proprietary multiplexers to aggregate traffic from different services, originally for the corporate networks market. Historically, these have been based on time division multiplexing (TDM), although cell relay and asynchronous transfer mode (ATM) based equipment is also in increasing commercial use. Multiplexer manufacturers include voice compression capability within their product ranges to ensure that the capability offered by a single leased line is maximised. In countries where public voice telephony has been deregulated, access to alternative service providers can be obtained via the incumbent's access network. An appropriate prefix is added to the dialled digits to signal to the incumbent voice switches that the call should be routed to an alternative service provider's network ('indirect

access'). This prefix can be entered manually by the customer or automatically by a PBX or 'smartbox' using a pre-assigned allocation of a prefix to classes of dialled digits; further details are given in Mallett and Fletcher [7].

## 5. The immediate future — telcos

In order to address the current issues affecting a typical telco access network, a range of commercial and technical solutions will be used to continue to drive down costs while simultaneously increasing the earning potential of the access business.

### 5.1 Costs

Cost reduction will be achieved by progressive reduction in capital investment in mature service delivery mechanisms. As competition increases, so too will churn between one operator and another, and new solutions will therefore be deployed specifically to control the cost of churn and the impact that churn traditionally has on network quality. The impact of overhead costs will be lessened by initiatives that spread the cost across an increasing volume of new services carried by the access network or by selective write-offs of obsolete networks.

### 5.2 Volumes

Volumes for mature delivery mechanisms (such as narrowband copper for POTS) will remain static in the face of growing retail competition; but there may be scope for growth in the wholesale market. Stock levels and just-in-time delivery processes will be optimised especially when providing connectivity to new greenfield sites. There will be a shift from a mainly reactive build process in the access network to a focused pro-active 'build-and-sell' strategy.

The imminent datawave will produce a growing demand for digital narrowband and broadband capacity both from existing customers and the emerging service provider market. Mobile networks will compete more with the traditional hardwired markets to provide narrowband speech service, but there will be an increasingly blurred boundary between fixed and mobile services and network infrastructure.

### 5.3 Capacity management

Access will continue to be predominantly via a hardwired network. Increasingly the copper network will be reserved for the provision of higher bandwidth services using DSL technology, and more extensive use will be made of pair gain devices (with up to six channels) to provide additional channels. Fibre will be used where it is cost effective to do so and where it enables high bandwidth services. A significant proportion of provision will be by

radio to manage churn and to reduce the need to install new copper or fibre cables.

### 5.4 Structural changes

Certain customers and market segments will require levels of reliability significantly better than average. Investment will be selectively applied to improve infrastructure in a targeted manner. In other cases where speed of provision becomes a crucial competitive advantage, pre-provision allowing high levels of 'met from stock' may well be justified. Less capital-intensive solutions will be needed for cases where the commercial value is less or where significant market share is unlikely.

The evolution of broadband services demands that the design approach to the first concentration point in the access network (broadband access node) is improved. Solutions are being developed to provide for short-term needs but with sufficient flexibility to evolve and support the datawave, thereby avoiding the risk of a proliferation of 'stovepipe' solutions. A similar approach to standardisation and simplification of network terminating equipment (NTE) for broadband and business products is being developed.

### 5.5 Operational costs

More of the network will be managed remotely by reducing the need for reactive interventions to provide service. Increasingly, flexibility will be provided by using radio and pair-gain devices. Improved reliability combined with a substantially lower requirement for flexibility will change the management of the access network from 'hands-on' to 'hands-off'.

### 5.6 Network health and reliability

Systems will be used to monitor the health of the network, adopt optimum stock levels, anticipate component failure and direct engineers to network components that require work. Network health scores will also become a key management performance measure. Maintaining accurate data and records is absolutely crucial to informed decision making and business success.

### 5.7 People and culture

The workforce will be organised in customer service teams trained, focused and incentivised to provide service at a cost and quality that will meet the commercial requirements of the business and provide exceptional customer service. They will function as separate businesses to exploit the commercial aspects of the local access network. This will also enable the development of more pride and ownership of the 'patch', which should be reflected in improved quality and customer satisfaction.

## 6. Telco evolution to broadband

There are then a bewildering number of choices for an existing telco to consider when upgrading the access network. One key element will be capital cost since this will have a major bearing on the commercial case for upgrade. Figure 11 compares the cost of potential broadband upgrade technologies as a function of penetration, expressed as a percentage of homes passed — and assumes that these are costs to an existing PTT with a copper access network already installed.

Not surprisingly therefore the DSL technologies have the lowest cost since they exploit the presence of copper pairs. FTTE represents ADSL technology in a fibre to the exchange architecture. FTTCab represents VDSL in a fibre to the cabinet architecture (Fig 8). HFC assumes the use of some existing BT duct but otherwise demands new infrastructure. FTTK and FTTH represent fibre to the kerb and home respectively.

Figure 11 clearly shows that ADSL is the lowest risk way of offering higher bandwidths since it is by far the lowest cost technology, particularly at the low penetrations that would be expected during the early phases of the market. FTTCab would similarly be the lowest risk way of offering higher bandwidths required for a full range of new interactive services, including broadcast TV.

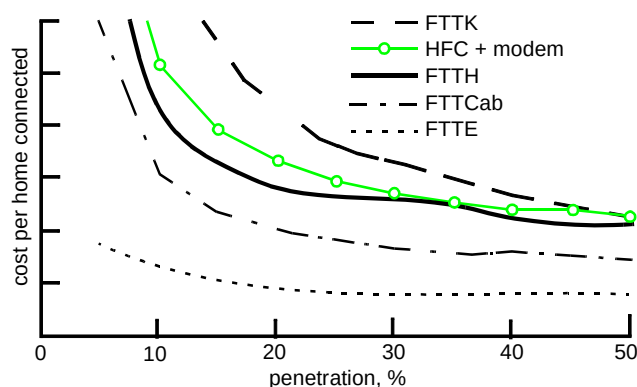


Fig 11 Cost versus penetration.

However, it should be recognised that it will take some time to roll out any broadband technology including ADSL, and possible migration steps are indicated in Table 4. Lowband is essentially ISDN technology and offers a digital service with 3—4 times the data capacity available to users of current voice-band modems. Midband might then be a next step offering the first big leap in bandwidths.

Highband could be introduced at a later date when demand for even greater bandwidth emerges and when the cost of the requisite technology becomes commensurate with potential network revenues.

Table 4 Broadband evolution steps.

|          |                    |                     |
|----------|--------------------|---------------------|
| Lowband  | ISDN2/Home Highway | 64—128 kbit/s       |
| Midband  | ADSL               | 500 kbit/s—6 Mbit/s |
| Highband | FTTCab/VDSL        | up to 25 Mbit/s     |

Indeed lowband is already being offered by many telcos as standard ISDN. BT has recently launched a new lowband product in the form of HomeHighway (Fig 12) which allows both PSTN and ISDN connectivity at the NTE.

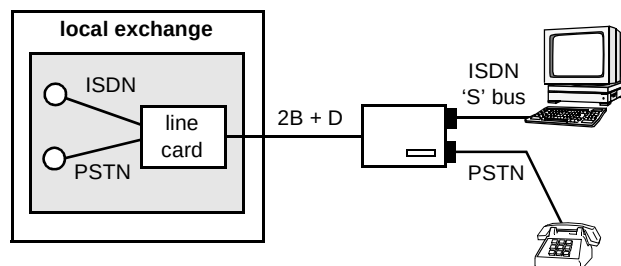


Fig 12 Lowband — ISDN HomeHighway.

Midband is currently the subject of many trials around the world and is detailed elsewhere [8]. Similarly there is much activity on Highband around the world largely in the form of the Full Service Access Network (FSAN) initiative, as described in Quayle et al [9].

## 7. Choice for other operators

The upgrade path for other existing operators will largely depend on their initial choice of installed technology. Table 5 shows some possible ways forward based on the information first given in Table 1.

Table 5 Access technology upgrade.

| Business                       | Possible upgrade                 |
|--------------------------------|----------------------------------|
| Mobile/niche telephony         | NB data/UMTS                     |
| Metropolitan/business services | Geographical coverage            |
| Cable TV/telephony             | Digital/cable modem              |
| Satellite TV                   | Interactive back channel         |
| Long distance                  | Direct access (owned and leased) |

The mobile and fixed radio operators will offer narrowband data connections and in the longer term may turn to UMTS (the next generation mobile standard). The MAN operators already offer broadband access through the use of direct fibre connection and will opt for increased geographical coverage, e.g. COLT. The cable TV companies will offer digital services and install cable modems for Internet services. Satellite TV will offer interactive back-channels using PSTN to enhance their

services. Finally the long-distance companies will move into the local market and direct access either through their own or leased infrastructure.

## 8. Future market

As we move into the future, the market structure will change as regulation allows more liberalisation and operators jostle for position. Figure 13 outlines the direction in which the market is moving. One key difference shows ISPs moving more towards content rather than network as regulatory changes allow telcos to provide IP networks as part of the regulated businesses. Telcos are shown having moved extensively into the data market and offering broadband access. Long-distance companies have moved into local services through the merger with cable companies (CWC). Although this gives a predominantly UK view there are also signs that the US market may follow similar trends.

## 9. Conclusions

In conclusion it has been shown that the telecommunications market is very dynamic at the moment with new and existing operators jostling for position. Factors affecting the market include regulation, which is encouraging new entrants into the market, as well as the advent of new interactive services which is placing new requirements on the network. In addition there is now a large choice of new access technologies becoming available, and this choice is a critical part of each operator's strategy since access forms a dominant part of the investment required as well as determining the level of service offered.

The choice may clearly be different between new entrant operators and established companies who may

already have an installed base of one type of network. In particular, a suitable choice for a telco would be the use of DSL technologies since they exploit the existing copper network. In addition the best way forward would be through low, mid and highband phases as the market develops over a period of time.

In a world where telecommunications continues to change at an increasing pace, the access business will need to re-invent itself. There are no easy or cheap options available for access and no single initiative will instantly transform an incumbent access network. Although the access network dates back to the earliest days of telecommunications, it remains critical in the delivery of even the most advanced and modern service to the user. Thus the future of access is assured but the structure of the access business and the underlying access technology will inevitably continue to evolve.

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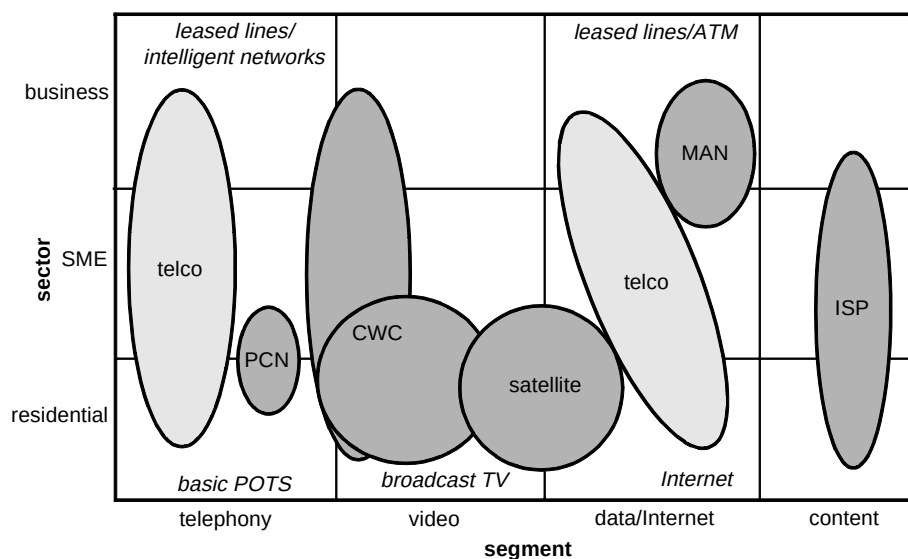


Fig 13 Future market structure

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Dave Spirit received a PhD in Physics in 1984 from Imperial College, London. Following two years as a researcher at Imperial College, he joined BT Laboratories in 1986 to work on the early development of optical fibre amplifiers. He spent a number of years in the area of optical transmission and networking, which included the first collaborative demonstration of 20 Gbit/s transmission in the BT network in 1994. In 1995, he joined a team built up to provide network design expertise for BT's Joint Venture companies. This has encompassed voice, data, access and core transport networks, particularly a pan-European network to be constructed by connecting JV networks together. He is a Member of the Institute of Physics and a Chartered Physicist, and is co-editor of 'High Capacity Optical Transmission Explained'.



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